

Optimized Personnel Tracking via Theoretical-Actual Acoustic Matching and Joint Active-Passive Constraint

Zhongxu Bao
China University of Mining and
Technology
Xuzhou, China
baozx@cumt.edu.cn

Chenxi Liu
China University of Mining and
Technology
Xuzhou, China
l_cx@cumt.edu.cn

Xuehan Zhang
Southeast University
Nanjing, China
xuehanzhang@seu.edu.cn

Xu Yang
China University of Mining and
Technology
Xuzhou, China
yang_xu@cumt.edu.cn

Yuqing Yin
China University of Mining and
Technology
Xuzhou, China
yinyuqing@cumt.edu.cn

Qiang Niu
China University of Mining and
Technology
Xuzhou, China
niuq@cumt.edu.cn

Abstract

Indoor personnel tracking plays a vital role in smart home applications such as security monitoring, elderly care, and appliance control. While passive acoustic localization offers privacy-preserving and non-invasive tracking by estimating the angle-of-arrival (AoA) of sound, it struggles in complex environments with low signal-to-noise ratios and multiple sound sources. Active acoustic methods improve robustness but often suffer from limited deployment flexibility and high computational cost. To address these issues, this paper proposes a novel acoustic signal fusion solution via utilizing audio system microphones, fully integrating active and passive acoustic signals. Our approach corrects angular prediction biases in passive sensing and incorporates beamforming-based ranging for accurate distance estimation. By comparing theoretical sampling matrices with real-time data and applying conjugate multiplication, we jointly optimize angle and range measurements. A fusion model further enhances overall tracking accuracy. Extensive experiments show that the proposed method achieves a tracking range of 5.2 m, reduces the median position error to under 11 cm, and improves angular accuracy compared to both passive-only and active-only baselines.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

acoustic waves, active and passive fusion positioning, beamforming, joint information searching.

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1 Introduction

Indoor personnel tracking[13, 17, 28] has attracted wide attention in the field of smart homes, supporting various applications such as security monitoring[4], elderly care[9], smart lighting[7], and home appliance control[38]. Meanwhile, contactless indoor personnel positioning[6, 22, 23, 34] based on acoustic signals has been widely applied, since it utilizes acoustic[8, 18, 25, 32, 33, 37] devices embedded in existing household smart appliances to sense, which can achieve strong privacy protection and non-invasive functions compared to wearable[24, 35] and camera[15, 20, 30] positioning methods.

Passive acoustic localization methods, which estimate the angle-of-arrival (AoA) of sound sources, are widely used to determine the location of individuals. However, their performance degrades significantly in complex environments [1, 3, 5] due to noise, low signal-to-noise ratios, multiple sound sources, and reverberation. Extensive research has focused on improving AoA estimation for passive acoustic localization, yet challenges in achieving high precision persist due to limitations in data dependency, synchronization issues, and complex environments [2, 19, 29].

To overcome the limitations of existing passive methods, recent researches[14, 26, 36] integrate passive sound localization with active visual sensing. While this fusion improves performance in certain scenarios, it causes high computational costs, especially in low-light or cluttered environments. In parallel with these developments, active acoustic perception methods have gained significant attention[11, 27, 31]. Compared to vision-based methods, active acoustic perception can break the boundaries such as fog, smoke, or thin obstacles, making it more robust for tracking and localization in complex environments. However, challenges remain,

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Corresponding authors: Xu Yang (yang_xu@cumt.edu.cn) and Yuqing Yin (yinyuqing@cumt.edu.cn).

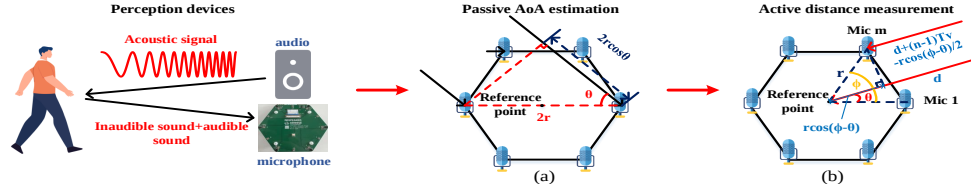


Figure 1: Hexagonal microphone array-based sensing models.

including high deployment costs, limited device deployment flexibility, restricted perception range, and reliance on model training with labeled data, which also causes performance degradation in unfamiliar environments [10, 12, 16, 21].

Based on above challenges, we propose a novel indoor personnel tracking approach that harnesses the full potential of active and passive acoustic signals. This approach not only corrects the angular prediction bias of passive signals but also enables accurate target ranging through active acoustic beamforming. To optimize both angle detection and position refinement, we compare theoretical sample matrices with actual collected values to solve the angle measurement and joint angle-ranging constraints. Finally, experimental results show that the proposed method achieves a human tracking range of 5.2 meters, with a median tracking error below 11 cm, outperforming both passive and active acoustic methods.

2 System Model

In this section, we consider using a hexagonal microphone array (which is actually a special case of a circular microphone array) as the benchmark for sensor modeling. We have developed models for angle estimation, distance measurement, and fine-grained position sensing. These models are applicable to any form of circular microphone array or its equivalent, with the specific methods outlined as follows.

2.1 Estimating the Target's Angle

AoA is an important feature for determining human position, indicating the direction of an acoustic signal reaching a microphone array. As shown in Fig. 1(a), sound waves travel varying distances to reach different microphones, allowing AoA calculation by analyzing these differences to reveal the acoustic source's orientation. Given the small microphone spacing (a few centimeters) relative to the detection region, acoustic wave trajectories are approximated as parallel lines. The propagation distance difference, denoted as Δd , and AoA θ determine the time difference of arrival (TDoA) for paired received signals.

$$\Delta \tau = \frac{\Delta d}{c} = \frac{r' \cos \theta}{c}, \quad (1)$$

where c is the sound propagation speed in air, and r' represents the distance between any two microphones. Traditional AoA calculation methods rely on estimating θ by $\Delta \tau$. Since microphone signals are discretely sampled, the limited time resolution constrains the precision of the estimated AOA θ , leading to quantization in angle and reduced spatial accuracy.

To improve the accuracy of θ , we achieve the most precise estimation of θ by minimizing the error between the observed and

predicted sample displacement differences. Thus, the complete optimization challenge can be expressed as

$$P1 : \min_{\{\theta\}} \left\{ \sum_{ij} |N_t(i, j, \theta) - N(i, j)| \right\} \quad (2)$$

$$s.t. \quad \frac{\pi}{9} \leq \theta \leq \frac{8\pi}{9}. \quad (3)$$

In (2), i and j represent the indices of the i -th and j -th microphones. $N_t(i, j, \theta)$ is the theoretical sample displacement difference for a given θ , while $N(i, j)$ is the actual displacement difference from the received signals. (3) represents the range of θ .

To preprocess the signal for AoA estimation, it is divided into 0.4-second windows to account for slow target movement. A Butterworth filter removes background noise outside the footstep frequency range. Signal displacement is then computed using the cross-correlation function between signals s_i and s_j , denoted as $R_{ij}(\Delta n)$. The sample displacement Δn represents the sample displacement difference between signals i and j . Given an array of M microphones, the sample displacement is

$$N(i, j) = \arg \max_{\Delta n} (R_{ij}(\Delta n)). \quad (4)$$

Thus, $N(i, j)$ forms an $M \times M$ matrix.

We then derive the theoretical sample displacement. As illustrated in Fig. 2, the signal propagation distance between a pair of microphones varies with the AoA, influencing the signal strength received. Given that the AoA of the target is θ , the theoretical sample displacement is computed as

$$N_t(i, j, \theta) = \frac{F_s \vec{\lambda}(\theta) \vec{P}_{ij}}{c \|\vec{\lambda}(\theta)\|}. \quad (5)$$

Here, $\vec{\lambda}(\theta)$ is the unit vector of the direction of the signal, and $\vec{P}_{ij}$ is the direction vector between the i -th and j -th microphones. When θ is fixed, $N_t(i, j)$ forms an $M \times M$ matrix.

By substituting $N(i, j)$ and $N_t(i, j, \theta)$ into (2), the theoretical value matrix closest to the actual value matrix corresponds to the estimated AoA.

2.2 Estimating the Target's Distance

This section presents a distance measurement model using a hexagonal microphone array for active acoustic sensing. By emitting a linearly varying chirp signal and analyzing its reflection from the human body, the model estimates the time of flight to determine distance. ToF is generally calculated from the intermediate frequency (IF) signal $S_m(t)$.

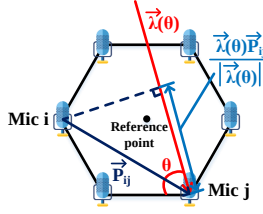


Figure 2: Model construction based on theoretical sample displacement differences from passive sound.

$$S_m(t) = \frac{1}{2}ae^{j2\pi\tau\left(\frac{B}{T}t+f_c\right)}. \quad (6)$$

where a denotes the signal amplitude; τ is the time delay; B is the bandwidth; T is the sweep duration; t is the time variable; and f_c is the central frequency.

The travel time of an acoustic signal is linked to human positional data, including distance, velocity, and orientation. As shown in Fig. 1(b), the microphone array's midpoint is the origin, with the target's distance to it denoted as d , yielding a travel time of $\frac{2d}{c}$. The target's velocity during the n -th chirp is v (negative when approaching), adding an extra travel distance of $(n-1)Tv$, which adjusts the travel time by $\frac{2(n-1)Tv}{c}$. The signal's AoA at the origin is θ , with the array consisting of M uniformly spaced microphones separated by r . The m -th microphone's angular position is $\phi = \frac{2\pi(m-1)}{M}$, resulting in a travel time difference of $\frac{r \cos(\phi - \theta)}{c}$ for the m -th microphone. Thus, the travel time τ for the n -th chirp at the m -th microphone is derived accordingly.

$$\tau(n, m) = \frac{2d + 2(n-1)Tv - r \cos\left(\frac{2\pi(m-1)}{M} - \theta\right)}{c}. \quad (7)$$

Substituting $\tau(n, m)$ into (6), the IF signal can be written as

$$S_M(t, m, n) = \frac{1}{2} a e^{j2\pi \left(\frac{B}{T} t + f_c \right) \frac{2d+2(n-1)Tv-r \cos\left(\frac{2\pi(m-1)}{M} - \theta\right)}{c}}. \quad (8)$$

The target's walking speed remains stable over short periods, and once the target's angle is determined, its distance can be calculated from the IF signal. By summing the signals from multiple microphones, the IF signal is enhanced. Additionally, noise and interference from other directions are minimized, thereby improving distance measurement accuracy and extending the sensing range. The IF signal optimization problem can be defined as

$$P2 : \max_{\{a, \theta\}} \left\{ \sum_{m=1}^M S_M(t, m, n) \right\} \quad (9)$$

$$s.t. \quad a \in \mathbb{C}, \quad (10)$$

$$\frac{\pi}{9} \leq \theta \leq \frac{8\pi}{9}. \quad (11)$$

The parameter a is a complex number to account for amplitude and phase adjustments in beamforming.

In AoA estimation, the footsteps signal’s AoA represents the human body’s angle relative to the microphone array. Using the array’s center as the reference, beamforming combines signals from

all microphones. The beamforming method involves substituting possible target angles $\hat{\theta}_s$ into (8). As human movement is negligible over short timeframes (ignoring speed), the IF signal for the m -th microphone is expressed accordingly.

$$S_M(t, m) = \frac{1}{2} a e^{j2\pi\left(\frac{B}{T}t + f_c\right)} \frac{2d - r \cos\left(\frac{2\pi(m-1)}{M} - \hat{\theta}_s\right)}{c}. \quad (12)$$

Since other microphones have phase differences relative to the reference point, each received signal is multiplied by a constructed compensation signal $S_{\text{add}}(t, m)$ to cancel these phase shifts.

$$S_{\text{add}}(t, m) = e^{j2\pi\left(\frac{B}{T}t + f_c\right)} r \cos\left(\frac{2\pi(m-1)}{M} - \hat{\theta}_s\right) \frac{2d}{c}, \quad (13)$$

and signals from all M microphones are summed to form the combined signal

$$S_{\text{new}}(t) = \sum_{m=1}^M (S_M(t, m) \times S_{\text{add}}(t, m)). \quad (14)$$

To account for environmental noise and the human body's width, beamforming is conducted at $\frac{\pi}{36}$ -radian intervals within a $\frac{\pi}{6}$ -radian range around the estimated AoA, selecting the IF signal with the highest energy. Applying a fast Fourier transform (FFT) to this signal extracts the frequency component f_{new} . Thus, the distance d to be calculated as

$$d = \frac{Tcf_{\text{new}}}{2B}, \quad (15)$$

where T , c , and B are known, and f_{new} corresponds to the largest frequency peak.

2.3 Trajectory Tracking Based on Joint Constraints

Considering that cluttered footstep sounds can affect AOA estimation results, it is necessary to further refine the previously obtained results.

As discussed, the IF signal carries critical information about the target’s distance, speed, and angle, but real-world detection environments introduce noise and multipath interference alongside human-reflected signals. Optimizing the IF signal is thus essential for enhancing positioning accuracy. To streamline extraction, three components are defined, specifically the distance el-

element $D(t, d) = e^{j2\pi\left(\frac{B}{T}t + f_c\right)\frac{2d}{c}}$, the speed element $V(t, n, v) = e^{j2\pi\left(\frac{B}{T}t + f_c\right)\frac{2(n-1)Tv}{c}}$, and the angle element and the angle element

$$\varphi(t, m, \theta) = e^{j2\pi\left(\frac{B}{T}t + f_c\right)\frac{r \cos\left(\frac{2\pi(m-1)}{M} - \theta\right)}{c}}.$$

Substituting three components into (8), the simplified intermediate frequency signal can be expressed as

$$S_M(t, m, n) = \frac{1}{2} a \frac{DV}{\varphi}. \quad (16)$$

To fully optimize the IF signal, we construct theoretical IF signal samples by iterating speed, angle, and distance. The closeness between the theoretical values $S_{MT}(t, m, n)$ and real values $S_M(t, m, n)$ is defined as

$$P3 : \min_{\{d,v,\theta\}} \left\{ \sum_{m=1}^M S_M(t, m, n) - \sum_{m=1}^M S_{MT}(t, m, n) \right\} \quad (17)$$

$$s.t. \quad 0 \leq d \leq 5, \quad (18)$$

$$-1 \leq v \leq 1, \quad (19)$$

$$\frac{\pi}{9} \leq \theta \leq \frac{8\pi}{9}. \quad (20)$$

Joint search relies on active sound; however, human reflections in the raw IF signal are often masked by background noise, reducing positioning accuracy. To address this, background subtraction is applied by removing the signal from the previous chirp cycle. This suppresses static and direct path components, leveraging target motion to isolate human reflections and enhance signal clarity.

The IF signal received by the microphone is primarily determined by the distance element, the velocity element, and the angular element, with the human body position defined as (d, v, θ) . Theoretical IF signal samples are generated by constructing all possible d, v, θ within the search range per (16) and compared with measured IF signals. The best-matching sample determines the estimated d, v, θ of the human body.

The search space for the position parameters (d, v, θ) is defined based on initial estimates $\mathbf{d}$ (distance) and Θ (angle) obtained from AoA and ranging results. Specifically, $d \in (\mathbf{d} - 0.2, \mathbf{d} + 0.2)$ m with a resolution of 0.01 m, $v \in (-1, 1)$ m/s with a resolution of 0.1 m/s, and $\theta \in (\Theta - \frac{\pi}{9}, \Theta + \frac{\pi}{9})$ with a resolution of $\frac{\pi}{90}$ rad. The measured IF signal within each search instance is denoted by $S(i, m, n)$, where $i = 1, \dots, I$ indexes time samples per chirp, $m = 1, \dots, M$ denotes the microphone index, and $n = 1, \dots, N$ represents the chirp index.

Next, construct the theoretical frequency signal samples. For all possible (d, v, θ) , the theoretical frequency signal sample can be represented as

$$S_t(i, m, n, d, v, \theta) = \frac{D_t(i, d)V_t(i, n, v)}{\partial_t(i, m, \theta)}. \quad (21)$$

To improve the robustness of signal matching under noise, we multiply the actual IF signal sample with the conjugate of the theoretical IF signal sample. This operation preserves key signal features such as amplitude while effectively suppressing phase noise. By comparing the measured signal with a set of theoretical signals corresponding to different position hypotheses (d, v, θ) , the approach identifies the hypothesis that yields the highest similarity. The one with the maximum response corresponds to the estimated target position, velocity, and angle.

3 Experimental Evaluation

The experimental setup consists of a commercial speaker (EARISE AL-202) for emitting acoustic signals and a commercial microphone array (ReSpeaker 6-mic Circular Array) for signal reception. Both devices are linked to a Raspberry Pi 3B, which communicates with a laptop (Dell Inspiron 7566) via a shared network. The Raspberry Pi manages both the transmission and reception of acoustic signals and forwards the recorded data to the laptop for further processing in MATLAB. The emitted chirp signal is characterized by its initial frequency, bandwidth, and duration, while the microphone array operates at a sampling rate of 48 kHz.

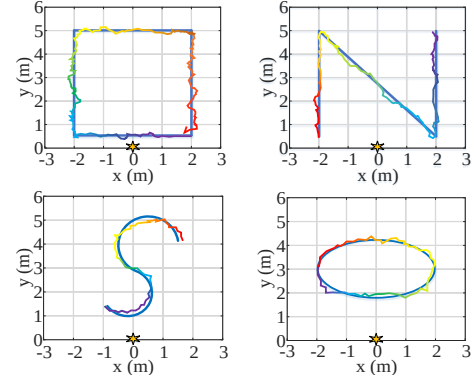


Figure 3: Trajectory tracking results.

To assess tracking accuracy, 10 volunteers walked along pre-defined routes under controlled conditions. Key positions were marked with tags, and a camera provided ground-truth data. The sensing range was 0–5 meters, with the transmitter placed at (0, 0) and 100 cm above ground. Fig. 3 illustrates the tracking results: yellow stars denote transceivers, the blue line is the reference path, and colored lines represent actual trajectories (red: start, purple: end). Volunteers walked naturally throughout the trials.

The proposed method achieves a median tracking error of less than 11 cm and extends the measurable range to 5.2 m, delivering substantial improvements in both positional and angular accuracy over passive-only and active-only baselines. Its performance was thoroughly evaluated across a range of factors, including environmental noise, individual differences, transceiver height, walking speed, and diverse real-world scenarios. Results show that the method is robust to ambient noise and maintains consistent accuracy across different individuals and motion patterns. It effectively tracks targets moving at speeds from 0.5 m/s to 1.5 m/s, demonstrating adaptability to dynamic conditions. While transceiver height has a minor effect, setting it at 1 m ensures stable performance. Experiments in environments such as parking lots, classrooms, large laboratories, and meeting rooms further confirm the method's robustness, scalability, and practical applicability in complex indoor settings.

4 Conclusion

This paper proposes a personnel tracking method based on the fusion of active and passive acoustic signals, aiming to exploit the complementary advantages of both modalities. Passive acoustic signals are used to estimate the angle of arrival via time-delay matrix matching, providing reliable directional information without the need for active transmission. Simultaneously, active acoustic signals facilitate distance measurement, with beamforming-based correction enhancing range accuracy. To further improve tracking precision, the method constructs theoretical IF signal samples to jointly constrain the target's speed, angle, and distance.

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